METHOD STATEMENT: WATER PROOFING

Client: Khalifa University Arzanah Campus

Contractor: White Crescent Décor

1.0 SCOPE

This Method Statement Outlines the procedures for the application of Waterproofing System. This method statement covers only the installation and details of the Waterproofing System.

2.0 PURPOSE

The purpose of this method statement is to define and describe the scope of work, control procedure and required inspections for the Waterproofing for System in accordance with the Project Specification relevant standards and contract documents

3.0 REFERENCES

Material Submittal for ,Waterproofing Works

4.0 DEFINITIONS

QA/QC – Quality Assurance and Quality Control
ITP – Inspection and Test Plan
WIR – Inspection Request
Supervision Consultant – M/s. STH Society Technology House LLC

5.0 RESPONSIBILITY

Project Manager / Construction Manager:

➢ The Project Manager / Construction Manager is responsible for the implementation of Method Statement.

➢ The Project Manager / Construction Manager shall ensure that the key personnel are aware of the Specification and Method Statement.

➢ The Project Manager / Construction Manager shall ensure that all resources like Manpower Material and Equipment are available to carry out the work as planned and without any delay or confusions.

Project Engineer:

➢ The Project Engineer is responsible to carry out the work as per the Method Statement waterproofing Specification and Approved Drawings.

➢ The Project Engineer shall ensure that the resources are available to carry out the works as scheduled.

➢ The Project Engineer shall be available at site full time for the execution of works as per schedule and specification.

QA/QC Engineer:

➢ The QA/QC Engineer is responsible for the effective monitoring implementation & documentation of QA/QC Procedures.
➢ The QA/QC Engineer shall ensure the implementation of ITP and Method Statements etc.
➢ The QA/QC Engineer shall ensure that all the activities are proceeding by the submission and approval of WIR and Check list.
➢ The QA/QC Engineer shall ensure that all the approvals are available for the safe movements of works.

Safety Officer:

➢ Safety Officer shall check the excavation equipment's on site and shall check the equipment certificates & operators' certificates.
➢ Safety Officer is responsible for the effective implementation of safety procedures.
➢ Safety Officer shall ensure that all safety precaution is in place and personnel on site are using the required protective equipment.
➢ Safety Officer shall conduct toolbox meeting before the execution of works.
➢ The Safety officer shall be available at site full time for the execution of works.
➢ The Safety officer shall implement full safety precautions for all works at night.

Foreman:

➢ The Foreman is responsible to carry out the work as per the Method Statement water proofing Specification and Approved Drawings.
The Foreman shall be available at site full time for the execution of works.
➢ Execute the works as per the instruction given by QA/QC & Site Engineers.
➢ Carry out works according to the approved Method Statement and materials.
➢ Inform Site Engineers of any activity ready for inspections.
➢ Provide daily report to Site Engineers.
➢ Make sure that safety is well practiced on site.
➢ Proper storage of materials.
➢ Control site disposal of waste materials.

Surveyor:

➢ The Surveyor should follow the zero level which it received from ADM.
➢ The Surveyor shall ensure that setting out carried out with theodolite, Total station, and dumpy level.

➢ The Surveyor shall ensure that all the survey related works carried out as per the approved drawings.

➢ For all survey related items he shall be available at site full time.

6.0 GENERAL REQUIREMENTS

6.1MATERIALS

➢ Materials for Waterproofing

1. Primer - SOPRADERE HT
2. Liquid Membrane – COLPHENE WA

6.2 MANPOWER

Manpower requirements such as Project Manager, Construction Manager, Project Engineer, Foreman, skilled labour, and helper will be available depend upon the site requirements.

6.3 EQUIPMENT

➢ Surveying Equipment

➢ Finishing tools/ hand tools (floats, trowels, levels, etc.)

➢ Wheel borrows

➢ Measuring tape

➢ Paint roller & paint brush

➢ Thinner or Solvent Cleaner

➢ Hand Gloves

➢ Scrapper

➢ PPE's (safety, helmet, jacket and shoes, eye protection, ear plug. Gloves, harness, first aid kit etc.)

6.4 TRAINING AND SUPERVISION

➢ Employees involved in application/installation of waterproofing shall be supervised by trained and competent Foreman / Site Engineers.

Construction Program:

➢ The works shall be carried out as per the approved sequence of program. General:

➢ All related documents, shop drawings, method of statements, material approvals, subcontractor's approvals and test certificates etc. must be approved and available upon request.

7.0 CONSTRUCTION PROGRAMME

The works shall be carried out as per the approved sequence of program.

8.0 PROCEDURE

8.1 GENERAL

➢ Before commencing any activity, the Engineer shall be notified.

It shall be ensuring that material, sample and method statement and waterproofing work accessories have been submitted as per the specification and approved by the Engineer.

➢ Waterproofing materials delivered to site shall be carefully unloaded & handled to prevent any damage and samples shall be sent for testing (as per Specification/ Municipality requirements).

➢ Before commencement of work the nature and location of all MEP openings and built-in items shall be ascertained.

8.2 SEQUENCE OF WORK)

Primer- SOPRADERE HT B) Liquid Membrane- COLPHENE WA

8.3 SOPRADERE PRIMER Weather Conditions:

➢ Application of SOPRADRE HT PRIMER range of membranes is not advisable of the ambient temperature is below 4-degree C. Application of membrane should be avoided in case of extreme weather conditions like a sandstorm or rains, etc. Substrate preparation:

➢ Substrate should be generally free of oil, grease and made dry. Any dew settled or other dampness should be thoroughly cleaned using a dry cloth. Honeycombs and spalled concrete to be repaired and all nailed heads and protrusions that are likely to puncture the membrane must be removed or ground. Any surface imperfections or cracks can be made good by using bond polymer modified crack filled.

Priming:

➢ Still the content of the thoroughly before application to remove all sediments.

➢ SOPRADERE HT may be applied by brush, roller, or squeegee. It should be applied on dry surface. The primer is applied at the coverage rate of 4-6m2/lt/coat.

➢ The coverage will depend on the porosity of the substrate. The primer should dry completely prior to the application of waterproofing membrane.

➢ Clean all tools immediately with solvent after use. Hardened materials can be removed mechanically.

8.4 LIQUID APPLIED MEMBRANE (COLPHENE WA)

Substrate preparation:

Attention to full and proper preparation of the substrate is essential for complete adhesion.

➢ All surfaces should be dry and free from contamination such as oil, grease, loose particles, decayed matter, moss, algal growth, laitance, and all traces of mould release oils and curing compounds. This is best achieved by lightly grit blasting or high-pressure water jetting the surface.

➢ Treatment with a proprietary biocide should be carried out after the grit blasting, in areas where moss, algae or similar growths have occurred.

➢ Spalled and deeply disintegrated concrete should be removed to sound concrete and repaired with a Suprema Repair system.

Application of COLPHENE WA

➢ Stir the contents of the drum thoroughly before application to remove all sediments.

➢ COLPHENE WA liquid applied membrane may be applied by brush, roller, or squeegee. It may be applied on damp surface but ensure that there is no water logging. The coating is normally applied in two coats.

➢ After the application of the first coat, apply the second coat at right angles to the first coat and only after the first coat has dried completely. On vertical areas, apply the coating in multiple layers to avoid sagging. Normal re-coat is12 hours. Make sure that the container is kept closed after each application. If cement sand render or plaster will be applied on top of the coating, then to a better key for bonding it is recommended to broadcast the coating with clean dry sand whilst the coating is still wet.

➢ Clean all tools immediately with water use. Hardened materials can be removed mechanically only.

➢ The Colphene WA should be mixed with clean, dry and chloride free sand at a 1:3 proportion.

Brush/Roller Trowel Application

➢ Apply the mixed material using a soft bristled brush or roller or trowel. The first coat should be well brushed into the substrate and be applied at w.f.t. of 1mm Finish the application one direction to give a neat appearance. Dampen the surface in case the brush begins to drag during application.

➢ The first coat should be allowed to cure for a period of 3 hours @ 350C and longer at lower temperature, prior to proceeding with the application of the second coat.

➢ The second coat also should be applied like the first coat at a w.f.t of 1mm, in one direction but does not require dampening of the surface.

➢ Use a finishing trowel for the best result, in the case where a roller is used to apply the material.

9.0 QA/QC

All material control and repairing shall be controlled as per the approved Project Quality Plan and Quality Control Procedure.

10.0 HOUSE KEEPING

All workplaces will be cleaned from waste materials and ensuring clean and healthy work environment as well as maintain free access and egress in the event of emergency

11.0 SAFETY

All works shall be controlled as per the Safety procedures and as per Manufacturers Material Safety Data Sheets (MSDS). Safety equipment will be always worn. Safety procedures are to be considered before, through and after the completion of the work to avoid any possibility of danger to humans or the construction.

Khalifa University Arzanah Campus Abu Dhabi

Company Name: White Crescent Décor Co.
Work Description: Risk assessment for Waterproofing works **Date: 18/07/2022**

Probability	Severity				
	Insignificant (1)	Minor (2)	Moderate (3)	Major (4)	Catastrophic (5)
Rare(1)	1	2	3	4	5
Possible (2)	2	4	6	8	10
Likely (3)	3	6	9	12	15
Often (4)	4	8	12	16	20
Frequent (5)	5	10	15	20	25

Risk Grading Matrix

15-25	Extreme Risk	Absolutely Unacceptably High Activity or process should not proceed in Current form.
8-12	High Risk	Unacceptably High. Activity or process should be modified to include remedial planning and be subject to detail EHS risk Assessment.
4-6	Moderate Risk	Acceptable but must be managed "As low as Reasonably practicable" [ALARP] Activity or process can operate subject to management and or modification.
1-3	Lower Risk	Acceptable without required further actions No action is required unless escalation of risk is possible

Hierarchy of Controls [Control Identified may be a mixture of the hierarchy in order to provide minimum operator espouser]:

Elimination	Eliminate The Hazard
Substitution	Provide an Alternative that is capable of performing same task and is safer to use
Engineering Controls	Acceptable but must be managed at {As low as reasonably practicable} ALARP Activity or process can operate subject to and or modification
Administrative Controls	Develop Policies Procedures practices and guide lines in consultation with employees to mitigate the risk. Provide training , instruction and supervision about the hazard
Personal Protective	Personal protective equipment design to protect the individual from the hazard

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Khalifa University Arzanah Campus Abu Dhabi

Equipment	

S/N	Activity Element	Significant Potential Hazard	Risk Classification			Risk level Low/ Medium / High	Action To Be taken To Reduce The Risk	Revised Risk Classification			Residual Risk Low/ Medium/ High	Reference
			(PR)	(S)	Risk rating P*S			(PR)	(S)	Risk Rating P*S		
1.	Water proofing Paint & Materials Delivery, Storage and Distributions safe access & egress to site	Storage of Material/ Plant and Equipment on Collapse/Collision	3	3	9	M	• Approved method statement must be in place prior to start the work activity • Safe storage of Paint & other materials as per the manufacture recommendation (Cold Storage –COSHH Room) • Palletized loads should not exceed two pallets. • Submit MSDS for all cladding materials & wooden pack. • Not to be stored in direct sunlight • All Sub-con personnel enter inside the site must attend the site safety induction • Proper & safe access & egress to each zones &work place	1	3	3	L	OSHAD cop 8.0, 01.0, 17.0 and 2.0.36.0 & 53.0
		Overstress & overexertion in manual handling	3	4	12	H	• Use mechanical aid for heavy loads • No manual lifting beyond 50kg loads • Provide proper training for manual handling • Minimize manual lifting whenever possible	1	2	2	L	OSHAD cop 8.0, 01.0, 17.0 and

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												2.0.36.0 & 53.0
		Toppling of loads	3	4	12	H	• Well organized warehouse or lay down yard • Observe proper stocking of materials at designated area • No overloading handling equipment	1	2	2	L	OSHAD cop 8.0, 01.0, 17.0 and 2.0.36.0 & 53.0
		Hazards posed by mechanical handling equipment, e.g. Boom Loader & forklift	3	4	12	H	• Qualified operators only • Provide banksman in congested areas • Equipment in good operating condition & equipped with necessary safety gadgets • Mechanical equipment 3rd party certified	1	3	3	L	
2	Water Proofing Painting	Emission of Solvents Toxic Chemicals	3	3	9	H	• Use proper PPE (gloves, proper respiratory mask & plastic apron.) • Skin in contact with Paints should be washed immediately. • Ensuring adequate ventilation in the working area, • Remove all people from the painting area • Unauthorized entry totally prohibited at paint work ongoing area • Administrative measures, like providing sufficient safety and health training, instructions and information to the workers, appropriate supervision, good personal hygiene and good housekeeping	1	3	3	L	OSHAD cop 8.0, 52.0, 2.0.,23.0 & 53.0

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							• Paint work ongoing area need proper barricading & suitable warning sign boards • Paint & related other mixers bulk storage not at all permitted inside the work place • Paint & other mixer area stored at designated COSHH room or paint container only • Paint storage area to maintain proper standard as per the federal law & OSHAD standard • Only authorized & qualified persons do the spray painting activity at site. • Wearing Proper PPE's, Respiratory Mask during the painting activity • Paint Fumes not affected to Public & co –workers • Use proper access and ladders • Platform must be cleaned and remove all loose material Follow working at height safety requirement					
3	Manual Handling	Stresses and Strains Slips, Trips and Falls Crushing or Trapping	4	3	12	H	• Ensure adequate information, instruction, training and supervision. Wherever possible, use mechanical aids to reduce manual handling • Ensure each lift is planned and assessed, avoid double handling of materials • Ensure travelled route is free from obstacles or hazards and has adequate lighting • Ensure individuals are capable of lifting or moving the load • Assess the load for size, shape, weight, and stability before it is moved. Never try to lift above your own capabilities. • Carry out a test lift to ensure the load can be moved	1	3	3	L	OSHAD-SF- COP 14.0

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							comfortably, check for sharp edges (if required wear gloves to protect hands). • If moving heavy or obscure loads, use more than one person (one person should give the lifting instruction). • Ensure correct handling techniques are used, such as keeping the load close to the body, keep feet shoulder width apart, and avoid twisting or stooping while carrying load. • Avoid carrying heavy or obscure loads up or down stairs. Never carry heavy loads up or down ladders.					
4	Activities in Open Areas /High Temperature	Heat stress – high ambient temperatures & humidity (especially in June, July & August, September)	3	4	12	H	• Comply with work/rest hours as per MOL orders • (12:00 – 3:00 pm rest) • Provide iced/cold water at site and work stations • Encourage workers to drink lots of fluids regularly • Provide glucose/water solution as required	1	4	4	L	OSHAD-SF- COP 11.0
		Heavy winds (sandstorms)	3	3	9	M	• Temporarily seek shelter / cover • Ensure all structures & materials are securely fixed / stable in place • Suspend lifting operations	1	4	4	L	OSHAD-SF- COP 11.0,53.0
		Heavy rain	4	3	12	H	• Temporarily suspend certain activities to avoid accidents	1	3	3	L	OSHAD-SF- COP 11.0,53.0
5	Manual Handling	Stresses and Strains Slips, Trips and Falls Crushing or Trapping	4	3	12	H	• Ensure adequate information, instruction, training and supervision. Wherever possible, use mechanical aids to reduce manual handling • Ensure each lift is planned and assessed, avoid double handling of materials • Ensure travelled route is free from obstacles or	1	3	3	L	OSHAD-SF- COP 14.0

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							hazards and has adequate lighting • Ensure individuals are capable of lifting or moving the load • Assess the load for size, shape, weight, and stability before it is moved. Never try to lift above your own capabilities. • Carry out a test lift to ensure the load can be moved comfortably, check for sharp edges (if required wear gloves to protect hands). • Ensure correct handling techniques are used, such as keeping the load close to the body, keep feet shoulder width apart, and avoid twisting or stooping while carrying load. • Avoid carrying heavy or obscure loads up or down stairs. Never carry heavy loads up or down ladders.					
6	**Storage & Use of Hazardous, Substances** **(Chemical, Paints,** **lubricants, sealants,** **solvents)**	Inhalation of Dust Fumes Vapors, vaporized metals - Contact with skin Contact with eyes Falls of	3	3	9	H	• Ensure Job safety analysis (JSA) is completed. • All substances must have accompanied Material Safety Data Sheet from Manufacturer. • Follow MSDS storage requirements. • Ensure correct labelling on containers • Provide separate storage locations, properly marked, proper access & egress • Review hazard operations with concerned persons • Use chemical protective clothing, boots, face shields, goggles, gloves • COSHH Assessment on substance required suitable to work. • PPE must be worn as per recommended on label often/drum.	1	3	3	L	OSHAD cop 01.0, and 8.0.& 53.0

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		materials Storage and handling of chemicals throughout the project on soil and near water sources					• First Aider on site with adequate first aid facilities. • All workers aware of hazards from COSHH assessment. • Ensure adequate ventilation to move or extract fumes, and respiratory equipment worn. • Storage of substances must be secure and of good construction with ventilation, placed in cool and shaded area and out of direct sunlight. • Ensure substance and protective equipment is disposed of in accordance with safety data Sheet. • Always clean hands and face thoroughly before eating, drinking or smoking. • Use Barrier creams and moisturizers. • Suitable firefighting equipment to be suitably located and marked. • Ensure Hazard warning signage is displayed.					
		Spills , environmental/ atmospheric hazards, fire	3	4	12	H	• Establish and implement chemical spill emergency response plan; effective response procedures • Provide suitable containment	1	3	3	L	OSHAD cop 01.0, and 8.0.& 53.0

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7	**Waste Management (chemical Empty Drums & other Materials) Handling)**	storage and disposal of various categories of waste Incorrect disposal.	3	4	12	H	• Routine inspections for waste segregation & disposal of chemicals & materials. • Empty drums properly collected from site & store at designated area or disposal as per the standard procedure • The site shall undergo a daily clean up including removal of general litter and construction waste • Waste shall be disposed of at designated disposal waste site approved by EAD. • No waste will be buried or disposed of on-site • Lessons drawn from the incident shall be communicated to construction crew through tool box talk.	1	3	3	L	OSHAD cop 8.0, 54.0, and 2.0. **Federal Environmental Law No 24 Of 1999** Construction Waste Management Plan - **CWMP**

Prepared by: Mr. Muhammad Umar Farooq
Designation: HSE Manager
Signature:

Approved by: Mr. Maher Abbas
Designation: PROJECT MANAGER
Signature:

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